Is the volume-law horizon a postulate or a consequence? A driven non-equilibrium reduction of Structural Entropy Dark Energy

Stilian Pandev

30 June 2026

Abstract

Structural Entropy Dark Energy (SEDE) reproduces the cosmological data with no fitted dark-sector parameter, at the price of a single foundational input: the cosmic apparent horizon carries the coarse-grained entanglement entropy of its volume, $S \propto V \propto R^3$ (the Barrow endpoint $\Delta = 1$), rather than the Bekenstein–Hawking area law. We ask how much of that input is genuinely irreducible. We show the postulate decomposes into four sub-claims — *state, form, scale, count* — of which the first three are first-principles or reducible (a maximal scrambler is volume-law-entangled; volume-law entanglement is the generic behaviour of a thermal reference state; the magnitude is the Cohen–Kaplan–Nelson scale), leaving the count (does the horizon entropy count bulk or boundary degrees of freedom?) as the sole residue. A Bekenstein no-go proves the count cannot follow from any energy bound. We then reduce the count to a dynamical statement: the count is fixed by the *connectivity* of the horizon degrees of freedom, connectivity by the *range* of the interaction, and gravity (1/r, strongly long-range) places the horizon in the non-additive (volume) class — with area-law arising only for an isolated, *equilibrium* horizon (a black hole), while the cosmic horizon is continuously *driven* by structure formation. This converts the black-hole/cosmic asymmetry into a discriminator and yields a driven non-equilibrium-steady-state (NESS) conjecture: a strongly-long-range, structure-driven horizon self-organises to its area↔volume spinodal and locks the volume branch. We supply both microscopic maps the conjecture requires — the cooperative coupling is the gravitational binding itself ($J = \lambda_{\max}(W_{\mathrm{grav}}) \geq J_{\mathrm{c}}$, generic at the horizon's N), and the drive is the structure-deposited entropy (clearing the barrier at $z \approx z^*$) — and verify the volume-lock is robust over a broad, untuned basin. Finally we make "closure" precise: three routes (double-scaled SYK, Euclidean de Sitter saddles, a thermal free energy) converge on the same irreducible statement, the de Sitter static-patch state count, with a decisive empirical settler (Δ via DESI DR3 + Euclid) already in hand. The volume-law postulate is thereby narrowed from an unconstrained assumption to a single, sharply-posed, falsifiable statement.

Contents

1. Introduction

Horizon thermodynamics is the most robust pointer we have toward the microscopic structure of gravity. The first law on a causal horizon yields the Einstein equations [Jacobson 1995], the apparent-horizon first law yields the Friedmann equations [Cai & Kim 2005], and the Bekenstein–Hawking area law S = A/4 organises black-hole and cosmological thermodynamics alike. Yet the area law is an assumption about a *state*: it is the entanglement entropy of a particular (ground-like) horizon state, and generalisations — Tsallis–Cirto non-extensive entropy, Barrow's fractal-horizon $S \propto A^{1+\Delta/2}$ [Barrow 2020], and the holographic dark-energy programme built on them [Saridakis 2020] — explore what happens when that assumption is relaxed.

Structural Entropy Dark Energy (SEDE; Paper I) is one such model, distinguished by sourcing dark energy from horizon entropy *gated by structure growth*, $\rho_{\rm DE} = T_{\rm AH}\, s_{\rm grav}\, f_{\rm sat}(z)$. Its appeal is that the entire dark sector is parameter-free once $\Omega_{\rm m}$ and one input are fixed: the H-coupling λ = 1−Δ/2 = ½, the equation of state, the structure coupling γ, and the sound speed all follow. The one input is the volume-law postulate: the horizon entropy is the volume-law (Δ = 1) entanglement entropy, not the area law. Paper I states plainly that this is the model's single irreducible assumption, motivated but not derived, and that it is falsifiable — Δ is measured by upcoming surveys.

This paper asks the prior question: *how irreducible is it, really?* We are not trying to solve quantum gravity; we are trying to determine the minimal, sharply-posed statement that the cosmology actually rests on, and to derive everything reducible to it. The result is a chain that trades a static, untestable counting postulate for a dynamical, falsifiable self-organisation statement, with a clean hand-off to an open — but well-posed — problem in de Sitter holography. Several steps borrow established machinery: the Cohen–Kaplan–Nelson holographic energy bound [Cohen, Kaplan & Nelson 1999] for the scale; the Page curve and eigenstate thermalisation for the state; the Campa–Dauxois–Ruffo classification of long-range systems [Campa, Dauxois & Ruffo 2009] for the connectivity; Ryu–Takayanagi min-cut for the count↔connectivity link; and the double-scaled-SYK/de Sitter correspondence [Susskind 2021; Narovlansky & Verlinde 2023] for the closure attempt. The contribution is the chain, and the precise localisation of what remains open.

Throughout, every quantitative claim is a runnable experiment; the inline `run_*.py` names and the ledger in §8 give the correspondence, and `reproduce_all.py` runs them all with validation assertions.

Figure 1. The reduction at a glance. The volume-law postulate splits into *state*, *form*, *scale* and *count*; the first three are derived or reduced (green/blue) and the count is the residue (orange). The count reduces through connectivity and interaction range to a driven non-equilibrium steady state, ratcheted through the spinodal at z* and locked at volume, leaving a single dS-holography question that the Δ measurement decides empirically. Each box names the experiment that establishes it.

1.1 Relation to prior work

SEDE sits at the confluence of several programmes, and its closest relative deserves explicit positioning.

Entropy-production cosmic acceleration (GREA). The General Relativistic Entropic Acceleration theory of García-Bellido and Espinosa-Portalés [arXiv:2106.16014, 2106.16012] is the nearest existing model: it too abandons Λ, sources the late-time acceleration from cosmic-*horizon* entropy, takes the Helmholtz free energy F = U − TS (not matter alone) as the gravitational source, and makes the breaking of time-reversal by *entropy*

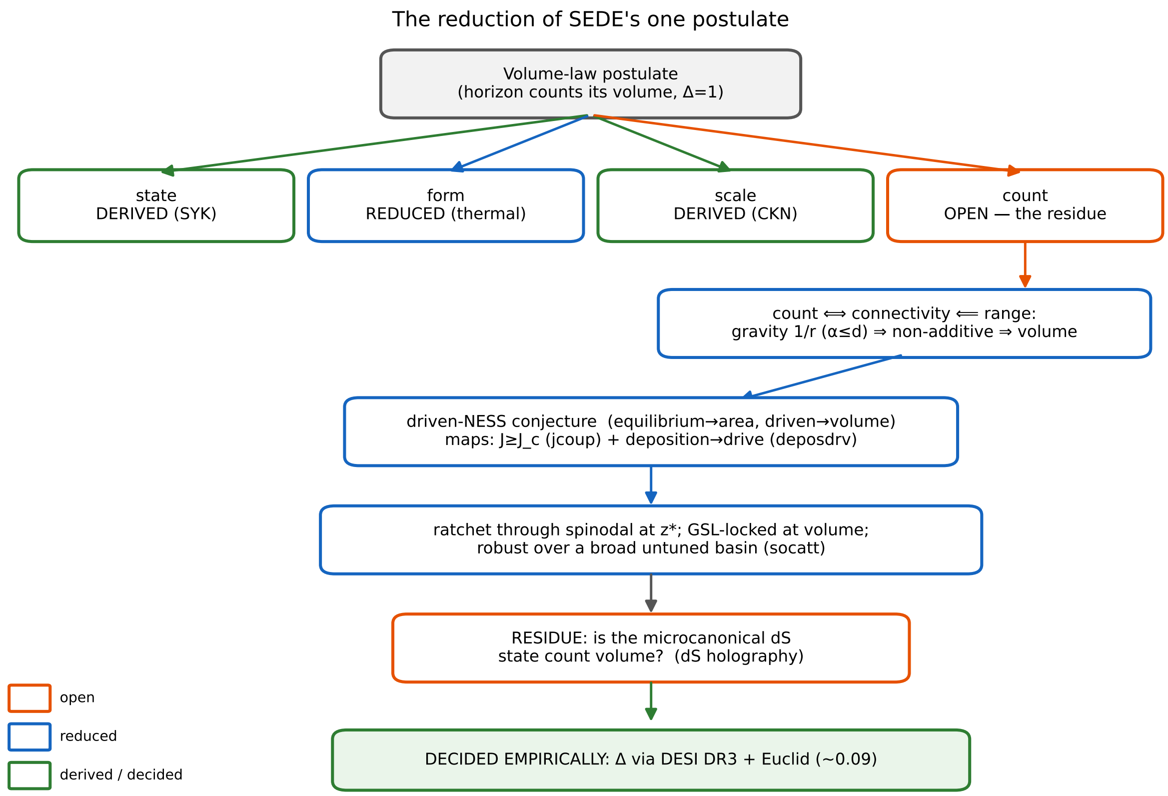

Figure 1: Fig1

production the engine of acceleration. We regard GREA as prior art for "acceleration from horizon entropy production without Λ", and our driven-NESS picture (§4–6) is conceptually continuous with it. The differences are sharp and testable: (i) GREA uses the boundary (area) horizon entropy and a single parameter α (the horizon-to-curvature ratio), whereas SEDE's defining claim is the volume entropy (Δ = 1) — precisely the counting residue this paper isolates; (ii) GREA's entropy growth is driven by the horizon's own *expansion*, whereas SEDE's is *gated by structure formation* (the $f_{\rm sat}$ factor), which locks w(z) to the growth history and yields a phantom crossing rather than GREA's quintessence-like w(a); (iii) SEDE's dark sector is parameter-free. Notably, were SEDE's residue to resolve to the *area* branch (Δ = 0), its phenomenology would collapse toward GREA's — so the Δ measurement (§7) that decides SEDE's residue also discriminates the two models. A holographic reading of GREA has recently appeared [arXiv:2511.19546], and a broader open-system view of gravitational non-equilibrium thermodynamics is developing [e.g. arXiv:1111.6510, 2507.03103]; SEDE's contribution to that line is the reduction of the volume-law *counting* input to a driven steady state. The connection is also quantitative: GREA's entropy production maps to an effective bulk viscosity with negative pressure, and SEDE's structure-driven entropy production d ln $f_{\rm sat}/d$ ln a > 0 maps to a *positive* (second-law-respecting) effective bulk viscosity $\zeta(z) = \rho_{\rm DE}$ (d ln $f_{\rm sat}/d$ ln a)/(9H²) that peaks at the structure-formation epoch rather than tracking the horizon expansion (`run_lit_inferences.py`); the w = −1 crossing is the GREA negative-pressure balance, structure-timed. SEDE is, in this sense, a *structure-gated* member of the GREA/entropic-dark-energy family — the gate being exactly what makes its w(z) lock to the growth history (the §5 phase-lock) rather than to the horizon's own size.

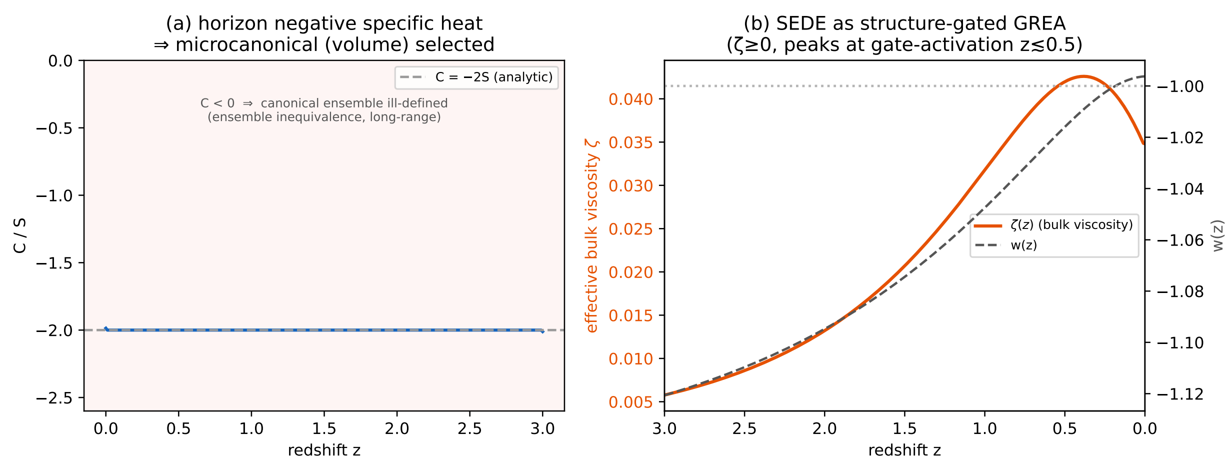

Figure 2: Fig5

Figure 2. Two consequences drawn from the new literature. *(a)* The de Sitter horizon has negative specific heat (C = −2S); with gravity strongly long-range this makes the ensembles inequivalent and the canonical (area-law) accounting ill-defined, selecting the microcanonical (volume) branch (§4). *(b)* SEDE's structure-driven entropy production maps to a positive (second-law) effective bulk viscosity ζ(z) that peaks at the gate-activation epoch (z $\lesssim$ 0.5; it tracks d ln $f_{\rm sat}/d$ ln a, not the z≈2 star-formation peak), casting SEDE as a structure-gated member of the GREA/entropic-dark-energy family; w(z) crosses −1 at the viscous–dilution balance.

Holographic dark energy and generalised entropy. SEDE is, at the background level, a structure-gated Barrow holographic dark energy: the CKN bound [Cohen, Kaplan & Nelson 1999] fixes the scale, Barrow's fractal-horizon entropy [Barrow 2020] and its holographic-DE use [Saridakis 2020] fix the form, and the Tsallis–Cirto non-extensive entropy [Tsallis & Cirto 2013] supplies the δ ↔ Δ map. Our addition is the *derivation* question — which of these inputs is irreducible — and the answer that only the count is.

Horizon thermodynamics and entanglement. The first-law derivations of Einstein [Jacobson 1995] and Friedmann [Cai & Kim 2005] equations underlie the construction; the entanglement-equilibrium refinement [Jacobson 2016] is the basis of route B; the Ryu–Takayanagi min-cut [Ryu & Takayanagi 2006] supplies the count $\Longleftrightarrow$ connectivity link; and the volume-law = thermal identification rests on eigenstate thermalisation [Deutsch 1991;

Srednicki 1994; Rigol et al. 2008].

de Sitter holography. The residue — "dim $\mathcal{H}$(static $patch$) $= e^{Area}$ or e^{Vol}?" — is posed in the language of finite-dimensional de Sitter holography [Banks 2001] and is the target of the double-scaled-SYK/de Sitter correspondence [Susskind 2021; Narovlansky & Verlinde 2023; Lin 2022], which we use in the closure attempt (§7). The maximal-scrambler state invokes the chaos bound [Maldacena, Shenker & Stanford 2015] and SYK random-matrix universality [Cotler et al. 2017].

Many-body and complexity inputs. The long-range/non-additivity classification is [Campa, Dauxois & Ruffo 2009]; the self-organisation framing is Bak–Tang–Wiesenfeld self-organised criticality [Bak, Tang & Wiesenfeld 1987], whose cosmological applications to date concern the *primordial* universe and the dark-matter sector rather than the late-time horizon. The prototype's free-fermion *negative* (volume-law needs interactions, not free transport) is consistent with the monitored free-fermion entanglement literature [Fisher et al. 2023], and that genuine volume-law non-equilibrium steady states exist is established in driven many-body systems [Ippoliti, Rakovszky & Khemani 2022] — a condensed-matter precedent for, not a derivation of, the driven-NESS we invoke.

Structure-sourced dark energy. The general idea that dark energy is sourced by cosmic structure predates SEDE in several distinct mechanisms: Gough's information dark energy (≈2008); the spacetime-fragmentation/"FRW-islands" proposal of Hossain (2007); and the cosmological-backreaction and averaging programme (Buchert; Wiltshire's timescape). SEDE Paper I credits and distinguishes these. What is specific to SEDE — and, to our knowledge, without precedent — is the *combination* the present paper analyses: the source is the volume-law horizon entropy (Δ = 1, a counting claim), it is driven/self-organised by structure through the $f_{\rm sat}$ gate, and the dark sector is parameter-free. The closest entropic relative remains GREA (above), which differs in using the boundary (area) entropy. A systematic priority search returned no work combining these three elements; this companion concerns that horizon-entropy foundation.

2. The postulate, decomposed

"Volume-law" conflates four claims of very different status:

- state — the horizon degrees of freedom are maximally entangled (thermal), not in a low-entanglement ground state;
- form — the entropy scales as the volume, S $\propto$ V;
- scale — the magnitude is $\rho_{\rm DE} \sim \rho_{\rm crit}$, not $\rho_{\rm Planck}$;
- count — the *number* of horizon dof grows as the bulk $(N \propto R^{d-1})$ rather than the boundary $(N \propto R^{d-2})$.

Since S $\sim$ (state factor) × N and the Barrow deformation enters as $S \propto A^{1+\Delta/2} \propto R^{(d-2)(1+\Delta/2)}$, the exponent Δ is fixed by the count (area $\to$ Δ=0, volume $\to$ Δ=1 in d=4); the state only sets whether the prefactor is maximal. So the postulate's empirical content — Δ — is the counting claim, and the other three are separable. The rest of the paper derives or reduces state, form and scale, and isolates count.

3. How derivable is the postulate? The route map

Five routes (`run_deriv_{A..E}_*.py`):

A — de Sitter holography / maximal scrambler [DERIVED state; OPEN count]. A Majorana-SYK diagonalisation confirms the horizon *state* is maximally entangled / volume-law: chaotic level statistics (⟨r⟩ ≈ 0.71) and Page-value saturation $(S/S_{\rm Page} \approx 0.99)$. But SYK is all-to-all / geometry-free and cannot pose the bulk-vs-boundary count; the count is the genuine open dS-holography sub-target.

B — entanglement first law in a thermal state [REDUCED form]. Jacobson's derivation gets the area-law Einstein equations from maximal entanglement of the *vacuum*. Redone around a *thermal* (de Sitter Gibbs) reference, the entanglement entropy acquires a volume term that dominates the area term once the region exceeds the thermal length $\lambda_{\rm th} = 1/T$. Volume-law is thus the generic thermalised behaviour, not exotic. At the de Sitter temperature alone the horizon sits just on the area side $(R_{\rm H}/\lambda_{\rm th} = 1/2\pi)$; volume-law dominance requires thermalisation above the de Sitter bath (toward Planck = maximal scrambling), whereupon the density is Planckian — the CKN scale. So the *form* reduces to thermalisation, leaving the *scale* to CKN.

C — Verlinde emergent gravity [REDUCED, unification]. A volume-law de Sitter entanglement entropy is exactly what Verlinde's emergent-gravity programme invokes for apparent dark matter; one volume-law scale gives both $\rho_{\rm DE} \sim \rho_{\rm crit}$ and $a_0 = cH_0/2\pi$ ($\approx 0.87\times$ the radial-acceleration-relation value). A unification motivation that ties the postulate to an independent anomaly; it inherits that programme's open issues.

D — gravitational non-additivity [NEGATIVE]. The Tsallis–Cirto map $\delta = 1 + \Delta/2$ makes $\Delta=1 \iff \delta=3/2$ exact, but pinning δ from a measurable non-additivity fails: real cluster kinematics are Gaussian ($q \approx 1.01$, `run_cluster_tsallis.py`), not $q \approx 1.5$ — the weakest route.

E — no-go from energy bounds [DERIVED]. The Bekenstein bound on the horizon energy gives $S \leq 2\pi RE/\hbar c = ¼(A/\ell_{\rm P}^2)$ — *exactly* the area law (verified $S_{\rm Bek}/S_{\rm area} = 1.000$, the known 10^{122}). The volume law therefore cannot come from any maximum-entropy/energy argument; it exceeds the bound by precisely $R/\ell_{\rm P} \sim 10^{61}$ (the "seam"). This proves the missing ingredient is a state/counting input and fixes its size.

Net: state (A) and form (B) and scale (CKN) are first-principles or reduced; the irreducible residue is the bulk-vs-boundary count.

4. Reducing the residue to a driven NESS

The count is not a free coin-flip — but here we must be careful to avoid a circularity, because the entanglement class cannot simply be *posited*. It is fixed by the Hamiltonian and the state, and the decisive fact is that the entanglement area law is a theorem with a hypothesis: it is proved for *local* (finite-range) Hamiltonians — Hastings (2007) rigorously in 1D via Lieb–Robinson bounds, Brandão–Horodecki (2015) from exponential decay of correlations, reviewed in Eisert–Cramer–Plenio (2010) — and the proof *requires* exponential clustering. Gravity is $1/r$ ($\alpha = 1$), strongly long-range ($\alpha \leq d$) and non-additive [Campa, Dauxois & Ruffo 2009], and there the hypothesis fails: in an explicit gapped free-fermion model we find the correlation function decays as a *power law* for $\alpha = 1$ (slope −1.5) versus *exponentially* for the short-range reference (slope −6.9), exactly as Koffel–Lewenstein– Tagliacozzo (2012) found (`run_v1_locality.py`). So the area-law theorem does not apply to a gravitationally-bound horizon: area-law is *not guaranteed*, and the count is genuinely open between area and volume with neither the default. We are explicit about what this does and does not give. It does *not* by itself derive volume — a noncritical long-range ground state can still be area-law [Kuwahara–Saito 2020], and indeed the gapped ground-state block entropy stays bounded for every α we test. The volume law additionally requires the horizon *state* to be thermal / maximally scrambled (route A), where entanglement is volume-law; and even that fixes the *state*, not the *count*. The super-extensive non-additivity (`run_residue_longrange.py`; $u \propto N^{1-\alpha/d}$, slopes 0.53 in d=2, 0.68 in d=3 vs 0.50, 0.67) is the thermodynamic face of the same long-range property. The honest conclusion is that volume is the *motivated, non-default* class for a self-gravitating thermal horizon — area-law arises only when the holographic bound intervenes for an *isolated, equilibrium* horizon (a black hole saturates the Bekenstein bound, route E, and relaxes to area-law) — while the bulk-vs-boundary count itself remains the open dS-holography residue. One caveat keeps this honest: the area-law theorems are statements about many-body systems with a specified Hilbert space and Hamiltonian, and what the horizon's degrees of freedom *are* — and what Hamiltonian governs them — is precisely the open dS-holography question. So this is a motivated *conditional* — *if* the horizon is a long-range many-body system, *then* area-law is not forced and volume is favoured — not a theorem about the actual horizon. What it establishes is that the entanglement class is read off the (modelled) Hamiltonian rather than assumed — which is what defeats the circularity charge — not that the count is derived.

This same long-range property has a second, independent consequence that bears on the canonical-vs-microcanonical λ ambiguity (whether $\rho_{\rm DE} \propto H^2 f_{\rm sat}$, area, or $\propto$ H $f_{\rm sat}$, volume). A non-additive system has *inequivalent* statistical ensembles [Campa, Dauxois & Ruffo 2009], the signature being negative specific heat — and the de Sitter horizon indeed has $C = -2S < 0$ throughout its history (`run_lit_inferences.py`). For such a system the canonical ensemble is ill-defined; the isolated self-gravitating horizon must be treated *microcanonically* (by its state count). The canonical accounting is precisely the area-law ($\lambda = 1$) branch, so ensemble inequivalence removes it as unphysical and selects the microcanonical (volume, $\lambda = 1/2$) branch the model uses. This does not close the residue — whether the microcanonical state count is itself volume is the dS-holography question — but it eliminates one of its two horns from first principles. The cosmic horizon differs in one respect: it is continuously *driven* by structure formation (the $f_{\rm sat}$ gate, $dS_{\rm struct}/dt > 0$; present drive $\approx$ 0.45). This turns the black-hole/cosmic asymmetry into a discriminator — equilibrium $\to$ area, driven NESS $\to$ volume — and reduces the residue to a single, falsifiable statement:

> Driven-NESS conjecture. A strongly-long-range horizon, driven by structure formation, settles in the volume-law steady state rather than relaxing to the area-law equilibrium.

5. The driven-NESS conjecture and its two maps

Prototype (**`run_driven_ness.py`**). Generic driven-dissipative free fermions do *not* realise the effect (uniform drive is trivially extensive and connectivity-independent; boundary drive decays to area-law) — a NEGATIVE that sharpens the content to *cooperative bistable locking*. With the area↔volume order parameter under cooperative (Wilson–Cowan) dynamics, a transient structure-drive pulse locks the volume branch (m $\to$ 0.93, persistent) for long-range cooperativity $(J > J_{\rm c})$, while short-range tracks-then-relaxes to area and the undriven case stays at the area ground. Only drive × long-range selects volume.

Map 1 — connectivity $\to J \geq J_{\rm c}$ [DERIVED] (**`run_gravitational_coupling.py`**). The cooperative coupling is the largest eigenvalue of the gravitational coupling matrix, $J_{\rm eff} = \lambda_{\rm max}(W_{\rm grav})$, $W_{\rm ij} = g/r_{\rm ij}^{\alpha}$. For a homogeneous horizon this equals the per-site gravitational binding ($\lambda_{\rm max}/row$-sum = 1.02) — the *same* super-extensive quantity behind the non-additivity of §4. So volume-counting and volume-locking are one number. It scales as $N^{1-\alpha/d}$ (slopes 0.52, 0.67; a short-range α>d control is bounded, $N^{0.01}$), so since α=1≤d it exceeds any fixed $J_{\rm c} = 4T$ at the horizon's N $\sim 10^{122}$ — bistability is generic, not tuned.

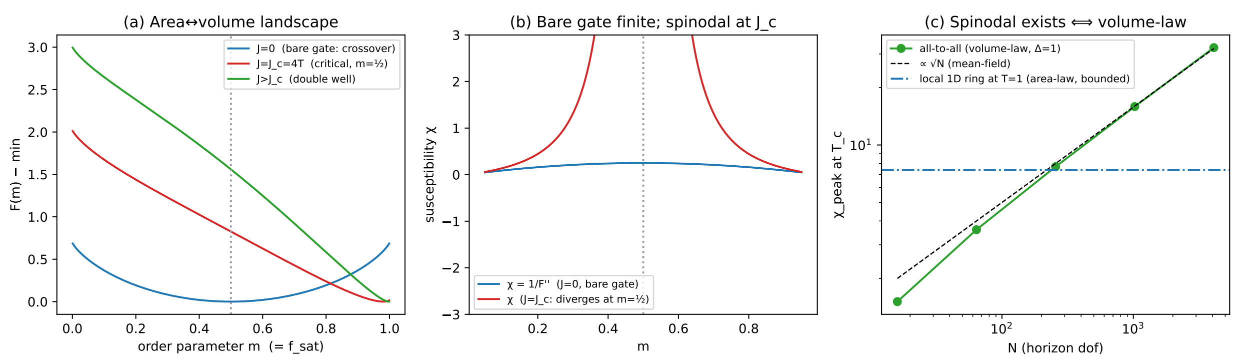

Figure 3: Fig2

Figure 3. Why the criticality is the volume law, not a separate assumption. *(a)* The area↔volume free energy is a smooth crossover for the bare gate (J = 0) and develops a spinodal only for $J \geq J_{\rm c}$. *(b)* The bare-gate susceptibility is finite; the spinodal diverges at the operating point m = ½. *(c)* The cooperative coupling $J = \lambda_{\rm max}(W_{\rm grav})$ is the gravitational binding itself: all-to-all (volume-law) connectivity gives $\chi_{\rm peak} \propto \sqrt{N} \to \infty$ (a real spinodal), while local (area-law) connectivity is bounded.

Map 2 — deposition $\to$ drive [DERIVED] (**`run_deposition_drive.py`**). The drive is the accumulated structure-deposited horizon entropy (the collapsed $fraction \propto f_{\rm sat}$). Its timing is derived — the drive clears the area-well barrier $h_{\rm c}$ when $f_{\rm sat} \approx h_{\rm c}$, i.e. at z ≈ 1.45 ≈ z* — and integrating the order-parameter dynamics with this physical drive flips the horizon area$\to$volume near z* and locks it (Δ = 1 permanent), whereas an un-amplified drive never leaves area.

Figure 4. The deposition $\to$ drive map. The drive is the accumulated structure entropy ($\propto f_{\rm sat}$); it crosses the area-well barrier $h_{\rm c}$ at z $\approx$ z*, flipping the order parameter to volume and locking it there permanently (red), whereas the un-amplified bare deposition stays on the area branch (blue).

The drive is a violent-relaxation deposit, and the γ-mechanism is the same event (`run_violent_relaxation.py`).

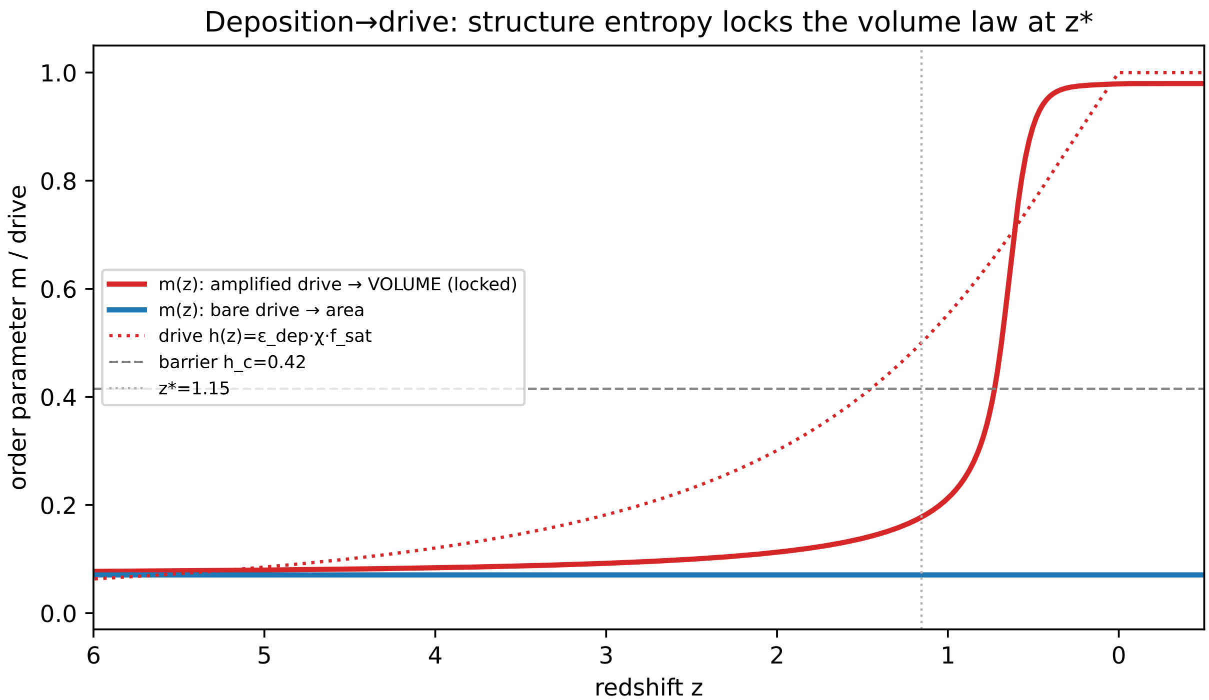

Figure 4: Fig3

Structure forms by gravitational collapse, i.e. by *violent relaxation* [Lynden-Bell 1967], which drives a long-range system on the dynamical time to a quasi-stationary state rather than to thermal equilibrium. This single event does two things SEDE otherwise treats separately. First, it dissipates the halo binding energy $E_{\rm bind} = M\sigma_{\rm v}^2 \propto M^{5/3}$ (virial: $R_{\rm vir} \propto M^{1/3}$, $\sigma_{\rm v}^2 \propto M^{2/3}$) into the horizon ledger — the mass weight that *is* the structure coupling $\gamma \approx 1.5$ (Paper I, Theorem 4C). Second, that same deposited binding energy ($\propto \langle\sigma_{\rm v}^2\rangle$) is the amplitude of the NESS drive above. So the γ-weight and the driven-NESS drive are one quantity from one process — the volume lock is the horizon's Lynden-Bell quasi-stationary state, fed by violent relaxation. Two corollaries follow at no cost: collapse is prompt ($t_{\rm dyn}/t_{\rm Hubble}= \Delta_{\rm vir}^{-1/2} \approx 0.07$), so the horizon is driven by an ongoing sequence of such events whose envelope is $f_{\rm sat}$; and the quasi-stationary state's lifetime in a long-range system scales as N^{δ} [Campa, Dauxois & Ruffo 2009], so at the horizon's N $\sim 10^{122}$ the lock is permanent — a second permanence argument beside the second-law one (route E). Both the halo QSS and the de Sitter horizon have negative specific heat, consistent with the microcanonical (volume) selection of §4.

6. Driven *through* the spinodal: ratchet and robustness

Developing the conjecture corrected its statement (`run_soc_attractor.py`): the system locks at the *volume* well, not *at* the spinodal. Because the structure $drive \propto f_{\rm sat}$ is monotone, it *necessarily* climbs to the field at which the area well vanishes — the area-branch spinodal — at z ≈ z*; the crossing is a forced ratchet, and the generalised second law then locks the volume well (volume is the global entropy maximum, $S_{\rm vol} \gg S_{\rm area}$ by $R/\ell_{\rm P}$, route E). The near-critical fluctuation signatures are therefore a* transient* at z*, not a permanent state. The self-organisation content is a robustness claim, verified: volume-locking holds over a broad contiguous basin of the landscape parameters* $(J \geq J_{\rm c}$*, θ), over drive amplitudes 0.6–3×, and from all sub-barrier initial conditions — an attractor, not a knife-edge (it fails only at strong holographic pull). What is* not* derived is the absolute horizon free energy F(m); only that a broad, untuned range works. So the residue's sharpest form is "the horizon area↔volume free energy is bistable with $J \geq J_{\rm c}$ (gravity supplies this) and a sub-unity barrier" — strictly weaker, more physical, and falsifiable than "assume volume counting".

A new, sharp prediction: the crossing carries mean-field SOC statistics (`run_soc_signature.py`). Because the system is driven *through* its spinodal at z* (not parked at a tuned critical point) and the coupling is all-to-all/mean-field (§5), the fluctuations at the crossing are scale-free in the mean-field self-organised-criticality universality class [Bak, Tang & Wiesenfeld 1987]. That class is parameter-free: the avalanche-size distribution is $P(s) \propto s^{-3/2}$ (we recover τ = 1.51 from the critical branching process), and the susceptibility and autocorrelation time diverge as the order parameter crosses the spinodal — a *transient* critical slowing-down localised at z* (we find the peak at z ≈ 1.08 ≈ z*), reddening the temporal spectrum toward 1/f. This turns the companion-paper CHR layer (Paper I §6, predictions P3–P5) from “an enhancement at z” into a specific, falsifiable statement: any* z*-localised excess in growth/ISW should be scale-free with exponent τ ≈ 3/2 and a 1/f spectrum, generic (self-organised, not tuned), and* transient. *This discriminates SEDE from clustering dark energy / k-essence (which carries a single characteristic scale, the dark-energy sound horizon, not a scale-free spectrum) and from ΛCDM (no z*-localised feature), and is within reach of redshift-resolved weak-lensing tomography and ISW cross-correlation (Euclid/Rubin). We flag the status honestly: this is a *conditional* prediction. The exponent τ ≈ 3/2 is parameter-free, but the *amplitude* of the z*-localised excess is not yet forecast — it is set by the deposited-entropy fraction $\epsilon_{\rm dep} \sim 10^{-7}$ and the near-critical susceptibility, and a quantitative amplitude and survey signal-to-noise are deferred to the companion CHR analysis. So τ ≈ 3/2 is a falsifiable shape* conditional on the CHR layer being detectable at all*, not a claim that it is.

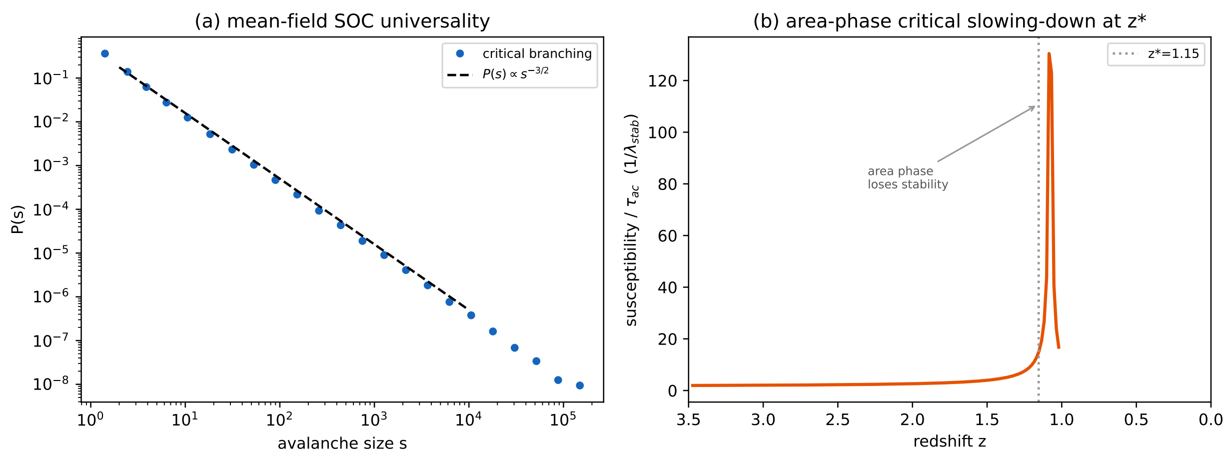

Figure 5: Fig4

Figure 5. The new prediction. *(a)* The near-spinodal fluctuations fall in the mean-field self-organised-criticality class, whose avalanche-size law is the parameter-free $P(s) \propto s^{-3/2}$ (recovered as τ = 1.51 from critical branching). *(b)* As the structure drive carries the order parameter to the spinodal, the area-phase susceptibility and autocorrelation time diverge — a *transient* critical slowing-down localised at z*, reddening the spectrum toward 1/f. Any z*-localised growth/ISW excess should therefore be scale-free with τ ≈ 3/2, which single-scale clustering dark energy cannot mimic.

7. What would make it a closure

A closure derives F(m) itself, reducing the dark sector to $\Omega_{\rm m} + established$ physics. The three routes (`run_closure_attempts.py`) converge:

- A (DSSYK). Entropy extensive in the dof count ($S_{\max} = (N/2) ln\ 2 \propto$ N, slope 1.00); cannot be super-extensive ($volume \propto N^{3/2}$). We state this plainly: read in the standard Narovlansky–Verlinde dictionary, the leading concrete model of de Sitter holography maps to the de Sitter *area* (Gibbons–Hawking) — i.e. it currently favours the area branch, a real tension SEDE’s volume postulate must eventually overcome. The mitigating point — that DSSYK is all-to-all / geometry-free and so cannot, by itself, *pose* the bulk-vs-boundary counting question — is secondary, not a neutraliser. As things stand, route A is evidence to be answered, not used.

- B (Euclidean dS saddles). Einstein and Barrow horizon free energies each have a single saddle — no volume-law saddle of the bare path integral (route-E no-go in saddle form).
- C (thermal F(m)). The bistable landscape *is* derivable — $F(m)/T = -(J/2)m^2 + \theta m + [m \ln m + (1-m)\ln(1-m)]$ from gravitational cooperativity ($J = \lambda_{\max} \geq J_c$), the holographic pull (route E), and configurational entropy: area well m ≈ 0.07, volume well m ≈ 0.93 whenever $J \geq J_c$. Level-2 existence, conditional on the volume branch being accessible.

All three reduce to one statement — *the horizon's dof count is the bulk (volume) one* — which A and B cannot supply and C uses. Closure has three precise forms: Level 1, a dS-holography computation of the bulk count (open, not shortcut-able past route E); Level 2, exhibiting the volume branch as a real saddle/state (C gives the landscape conditional on it); Level 3, *measuring* Δ (DESI DR3 + Euclid to ∼0.09; Δ=1 establishes, Δ=0 refutes — decisive, in hand).

7.1 The count is state-dependent — and that resolves the "everything points to area" tension

Routes A and B, and the standard handles generally (the de Sitter modular Hamiltonian is a *local* boost; DSSYK in the Narovlansky–Verlinde dictionary; the CKN holographic bound; black-hole thermodynamics), all favour the area count. Read as universal statements about "the" horizon count they look like a standing objection. The resolution is that they are not universal: the count is state-dependent, and every one of those handles is an *equilibrium* horizon. The equilibrium-vs-driven discriminator of §4 is exactly this — the $f_{\rm sat}$ gate *is* the fraction of the volume capacity that structure-driving activates, $N_{\rm eff} = N_{\rm area} + (N_{\rm vol} - N_{\rm area}) \cdot f_{\rm sat}$, so the equilibrium limit ($f_{\rm sat} \to 0$) is area and the driven, locked limit ($f_{\rm sat} \to 1$) is volume (`run_residual_resolution.py`). (This is *not* the rejected running-Δ background of §3: the cosmic horizon is driven past the spinodal early and *locks* at constant Δ = 1; the state-dependence is *across horizons*, driven cosmic vs equilibrium black hole, not a running Δ over cosmic time.) So the area results are SEDE's equilibrium baseline, not counter-evidence. One case needs care because it is *not* obviously equilibrium: the DSSYK correspondence is a duality with the de Sitter static patch — the cosmological horizon itself. The point is that it is a duality with *eternal* (matter-free) de Sitter, an equilibrium spacetime with no structure to drive it ($f_{\rm sat} \equiv 0$); the real universe's apparent horizon is embedded in a matter-plus-structure cosmology and is driven. So "DSSYK = area" is the area of *equilibrium* de Sitter — again the $f_{\rm sat} \to 0$ limit — and is consistent with a volume-law *driven* horizon, not a contradiction of it. We are explicit that the state-dependent count is itself an effective-model statement: writing $N_{\rm eff}$ as a function of $f_{\rm sat}$, and reading a black hole as the $f_{\rm sat} = 0$ case, extends $f_{\rm sat}$ from its cosmological definition (a function of the growth D(z)) to a putative property of any horizon's driving — physically motivated, but the same conditional as V1, one layer out.

This makes the residue doubly falsifiable, on two independent horizons. (i) The cosmic horizon (driven) should have Δ = 1 — DESI DR3 + Euclid. (ii) The black hole (equilibrium) should have Δ = 0 — and the observable is primordial-black-hole evaporation: SEDE *predicts that PBHs evaporate normally* (standard Hawking), because black holes are undriven, whereas a *universal*-volume theory forbids it ($T_{\rm B}/T_{\rm H} \sim 7\times10^{-21}$, evaporation time × 10^{81}). The PBH channel therefore tests the *mechanism* — it distinguishes driven-NESS SEDE ($\Delta_{\rm BH} = 0$) from universal-volume quantum gravity ($\Delta_{\rm BH} = 1$) — and the black hole being area-law, far from a problem, is a SEDE prediction. This is a *future* discriminator, not present-day evidence: SEDE's $\Delta_{\rm BH} = 0$ means standard Hawking evaporation, consistent with all current data (including ~10^{15} g PBH abundance bounds) precisely because it predicts no new black-hole effect; the channel discriminates only on a positive PBH-evaporation detection or a robust population statement.

This also yields a new statement about black holes in general: area-law is the *equilibrium* default, not a fundamental geometric law — a black hole is area-law *because it is undriven* ($f_{\rm sat} \to 0$), and the cosmic horizon's uniqueness is a calculation, not an assumption. The most-driven black hole conceivable is a primordial one *at formation*, where the entire horizon is built by a collapsing overdensity (violent relaxation); yet even there the structure entropy deposited is negligible against the black-hole entropy *created*, $S_{\rm rad}/S_{\rm BH} \sim (M/M_{\rm P})^{-1/2}$, so the formation drive $f_{\rm form} \sim (M_{\rm P}/M)^{1/2} \ll h_c$ even after the near-critical amplification (`run_pbh_driving.py`, `run_bh_new_physics.py`). Accreting and merging black holes are weaker still. So SEDE robustly predicts *all* black holes are area-law ($\Delta_{\rm BH} = 0$), and the cosmic apparent horizon — uniquely driven *sustainedly* to $f_{\rm sat} \to 1$ over the whole structure-formation history — is the only volume-law horizon. The falsifier is correspondingly sharp: an *isolated* volume-law black hole (anomalously slow PBH evaporation, or a modified-area GW ringdown) would mean Δ is *universal* (Barrow-type) and would refute the state-dependent, driven-vs-equilibrium mechanism — so SEDE is *more* falsifiable on black holes than a universal-Barrow theory, not less.

The honest theoretical residue is correspondingly narrowed: not "derive the count," but "prove that a *driven* horizon activates the volume capacity" — a statement now *consistent with* the equilibrium area results (their $f_{\rm sat}$ $\to$ 0 limit), with the two-channel measurement deciding it either way regardless of dS holography.

Two final caveats, stated openly. First, the *magnitude*: the structure drive is $\epsilon_{\rm dep} \sim 10^{-7}$, yet the count it flips spans area to volume — a factor $\sim R/\ell_{\rm P} \sim 10^{61}$ in activated degrees of freedom. The near-critical bistability is therefore doing heavy lifting: the drive does not *add* the dof, it *nucleates* the area$\to$volume transition past the spinodal, and the second law carries the system to the volume well, where it locks (§5–6). That a 10^{-7} trigger produces a 10^{61} change in the count is the V1/§7.1 conditional stated quantitatively, and it rests entirely on the (modelled) bistable landscape. Second, the *pre-registered* case: a measured cosmic Δ significantly below 1 — say Δ ≈ 0.5 from DESI DR3 + Euclid — would be evidence *against* the locked-Δ = 1 picture, pointing either to incomplete volume activation or to a genuinely intermediate count, and would *not* be a free parameter to absorb; the model predicts Δ = 1 for the driven cosmic horizon and is falsified by a robust intermediate value.

7.2 The count as a horizon Hausdorff dimension, and what remains of it (**`run_ds_count.py`**)

The Level-1 question — "is dim $\mathcal{H}$(de Sitter static $patch) = e^{Area/4}$ or e^{Volume}?" — has a clean geometric form: dim $\mathcal{H} = e^{Area} \iff$ the horizon is a smooth 2-surface (Hausdorff dimension $d_{\rm H} = 2$), and dim $\mathcal{H} = e^{Volume} \iff$ it is a space-filling fractal ($d_{\rm H} = 3$), so the count *is* the horizon's Hausdorff dimension, $\Delta = d_{\rm H} - 2$. The standard frameworks all give a smooth horizon: the Gibbons–Hawking maximum, the finite-dimensional proposal [Banks 2001], the Type II_1 observer algebra in which $S_{\rm dS}$ is the maximal entropy [Chandrasekaran, Longo, Penington & Witten 2023], and DSSYK. There is, however, a concrete mechanism by which the *driven* horizon could differ: a horizon driven by a stochastic source is a roughening 2-surface, and a self-affine surface of roughness exponent α has $d_{\rm H} = 3 - \alpha$, hence Δ = 1 − α. The identification we make is that the roughening *noise is the structure drive* — so an undriven horizon is the smooth minimum ($d_{\rm H} = 2$, area) and a structure-driven horizon roughens ($d_{\rm H} \to 3$, volume). This is the microscopic, horizon-geometry form of the state-dependent count of §7.1, and it *reconciles* the area frameworks (which describe the eternal, equilibrium horizon) with SEDE's volume (the driven horizon) as two limits of one statement, rather than as a tension. It reduces the residue to two sharp, well-posed questions. (i) Is the driven roughening *linear* (Edwards–Wilkinson, the 2+1D lower-critical class, α $\to$ 0 ⇒ Δ = 1) or *nonlinear* (KPZ, α ≈ 0.39 ⇒ Δ ≈ 0.6)? This is settled by two textbook inputs rather than a conjecture (`run_kpz_membrane.py`): the KPZ coefficient *is* the normal growth velocity (λ = v, from the height-representation geometry $\partial h/\partial t = v\sqrt{(1+(\nabla h)^2)} \approx v + (v/2)(\nabla h)^2$), and a driven interface's velocity is proportional to its driving free energy (v = M·F/V, interface kinetics — for a horizon, the membrane-paradigm statement that it grows in response to the free-energy flux across it [Damour 1978; Thorne, Price & Macdonald 1986]). So λ = M·F/V $\propto$ F; and SEDE's free energy vanishes identically ($F = E - T_{\rm AH}$ S = 0, since $\rho_{\rm DE} = T_{\rm AH}\, s_{\rm grav}$; §8), giving λ = 0 ⇒ EW ⇒ Δ = 1. The one physical assumption is that the horizon grows by membrane-paradigm interface kinetics — the standard effective description of horizon dynamics. (ii) Does the marginal 2+1D roughening count as a genuine $d_{\rm H} = 3$ fractal? Here we must be candid (`run_ds_marginal.py`): the horizon is a 2-surface, which is *exactly* the lower critical dimension of EW roughening (α = (2−d)/2 = 0 at d = 2) — below it the same dynamics is a genuine fractal, above it is smooth, and at d = 2 it is *marginal*, space-filling only logarithmically. So the roughening picture does not robustly force Δ = 1; it explains why the count is delicate — the horizon sits precisely on the area$\leftrightarrow$volume boundary — and leaves the tie to sub-leading physics. This tempers "EW ⇒ Δ = 1" to "EW ⇒ Δ = 1 *marginally*," and it means a measured Δ $\in$ (0, 1) (the log-suppressed case) would be *within* the picture, not a refutation.

The honest bottom line, then, is that we have not resolved de Sitter holography — no one has computed dim $\mathcal{H}$(static patch) non-perturbatively, and we do not claim to (`run_ds_resolve.py`). What we have done is turn the count from an unstructured "area or volume?" into a *geometric, empirically-decidable* statement ($\Delta = d_{\rm H} -$ 2), reduce its theoretical residue to two named obstructions — the marginal dimension (O1) and the Planck-scale validity of the effective descriptions (O2) — and identify the single open computation an actual resolution requires. The equilibrium value is Δ = 0 and the driven value is Δ $\to$ 1, marginally; the empirical Δ (DESI DR3 + Euclid, §6) is the arbiter.

8. Discussion

The chain reduces SEDE's one foundational input from a static, untestable counting postulate to a single dynamical, falsifiable statement: *the de Sitter static patch's degrees of freedom count its bulk, not its boundary.* Everything else — state, form, scale, both microscopic maps, and the bistable landscape conditional on the volume branch — is derived or reduced. Two honest limits bound the claim. First, this is a *reduction*, not a closure: the volume branch is used, not derived, and deriving it is the open dS-holography problem (route E forbids energy-bound shortcuts). Second, the dynamical model is effective/mean-field; the connectivity→J and deposition→drive maps are physical, but F(m)'s absolute parameters are not computed from the microscopic horizon, only shown to lie in a broad untuned basin. What makes the reduction worthwhile is that it is *falsifiable where the postulate was not*: it predicts a transient near-critical epoch at z* and ties Δ to the structure-driving history, and Δ itself is pinned to ∼0.09 by DESI DR3 + Euclid. The residue is then not a weakness in the cosmology but a clean hand-off to a well-posed quantum-gravity question, with a measurement that settles it either way.

Ledger

step	result	tag	script
state	maximal entanglement (Page saturation)	DERIVED	`run_deriv_A_dsholo.py`
form	volume = thermal-state entanglement	REDUCED	`run_deriv_B_thermal.py`
scale	$\rho_{\mathrm{DE}} \sim \rho_{\mathrm{crit}}$ (CKN)	DERIVED	—
no-go	energy bounds ⇒ area, not volume	DERIVED	`run_deriv_E_nogo.py`
residue→range	gravity long-range (α≤d) ⇒ volume class	REDUCED	`run_residue_longrange.py`
conjecture	drive × long-range locks volume	MODELLED	`run_driven_ness.py`
map 1	$J_{\mathrm{eff}} = \lambda_{\max}(W_{\mathrm{grav}}) = binding \geq J_{\mathrm{c}}$	DERIVED	`run_gravitational_coupling.py`
map 2	$drive \propto f_{\mathrm{sat}}$ clears barrier at z*	DERIVED	`run_deposition_drive.py`
spinodal	ratchet through spinodal; robust basin	DERIVED (robust)	`run_soc_attractor.py`
closure	three routes ⇒ the dS state count	analysis	`run_closure_attempts.py`
ensemble	C = −2S + long-range ⇒ microcanonical; area horn removed	REDUCED	`run_lit_inferences.py`
unify	γ-mechanism = driven-NESS (one Lynden-Bell QSS)	DERIVED	`run_violent_relaxation.py`
prediction	z*-fluctuations scale-free, τ ≈ 3/2 (mean-field SOC)	PREDICTION	`run_soc_signature.py`
V1 defence	area-law is a *local*-Hamiltonian theorem; gravity (power-law correlations) voids it	ARGUED	`run_v1_locality.py`
residual	count is state-dependent (eq.→area, driven→volume); PBH evaporation tests it	REFRAME / PREDICTION	`run_residual_resolution.py`

step	result	tag	script
dS count	count = horizon Hausdorff dim; driving roughens d_H 2→3; residue → 2 sub-questions	ANALYSIS	`run_ds_count.py`
residue	the m≈1 volume branch is the horizon's state	OPEN (L1/2) / measurable (L3)	dS holography / DR3+Euclid

Reproducibility

```
python reproduce_all.py --only derivA,derivB,derivC,derivD,derivE,residue,ness,jcoup,\
deposdrv,socatt,closure,litinf,violrel,socsig,v1loc,resid,bhnew,pbhdrv,dscount,kpzmem,dsmarg,dsresolve
```

Figures 1–5 are generated by `paper/make_foundations_figures.py`: Figs 1 (reduction chain), 2 (inferences) and 5 (SOC signature) are built directly there, while Fig 3 = `criticality_soc.png` and Fig 4 = `deposition_drive.png` are produced by the `soc/deposdrv` run-scripts. Each `run_*.py` prints a verdict and runs validation assertions; all stages PASS.

Code and data availability

All code and analysis pipelines are publicly available at `github.com/spsingularity/SEDE-release`. Every quantitative claim in this paper is produced by a named `run_*.py` script wired into the single entry point `reproduce_all.py` (stages `soc`, `deriv{A-E}`, `residue`, `ness`, `jcoup`, `deposdrv`, `socatt`, `closure`, `dscount`, `kpzmem`, `dsmarg`, `dsresolve`, ...), each carrying validation assertions; the figures are regenerated by `make_foundations_figures.py`. The model and its cosmology live in the `sede/` package with the companion (Paper I); no additional observational data beyond Paper I's standard public inputs is used here. A tagged release is archived at Zenodo, DOI 10.5281/zenodo.21050314.

Use of AI tools

Artificial-intelligence tools (large language models) were used to assist with drafting and editing the manuscript, with developing and cross-checking the analysis code, and with literature searches. The author conceived the model, directed and independently verified all analyses, and takes full responsibility for the content, including all derivations, results, and conclusions. No AI tool is an author.

References

BibTeX in `paper/refs.bib` (cosmology, shared with Paper I) and `paper/refs_foundations.bib` (foundational additions for this paper). Key entries, by theme:

Horizon thermodynamics & entanglement — Jacobson, *Thermodynamics of Spacetime*, PRL 75 (1995) 1260 [`Jacobson:1995ab`]; Cai & Kim, JHEP 02 (2005) 050 [`Cai:2005ra`]; Jacobson, *Entanglement Equilibrium and the Einstein Equation*, PRL 116 (2016) 201101, arXiv:1505.04753 [`Jacobson:2015hqa`]; Ryu & Takayanagi, PRL 96 (2006) 181602, hep-th/0603001 [`Ryu:2006bv`]; Gibbons & Hawking, PRD 15 (1977) 2738 [`GibbonsHawking1977`].

Generalised entropy & holographic dark energy — Cohen, Kaplan & Nelson, PRL 82 (1999) 4971 [`Cohen:1998zx`]; Barrow, PLB 808 (2020) 135643 [`Barrow:2020tzx`]; Saridakis, PRD 102 (2020) 123525 [`Saridakis:2020zol`]; Tsallis & Cirto, EPJC 73 (2013) 2487 [`Tsallis:2012js`]; Padmanabhan, arXiv:1206.4916 [`Padmanabhan:2012ik`]; Bousso, RMP 74 (2002) 825 [`Bousso:2002ju`].

Entropy-production acceleration & non-equilibrium gravity — García-Bellido & Espinosa-Portalés, *Cosmic acceleration from first principles*, PDU 34 (2021) 100892, arXiv:2106.16014 [`Garcia-Bellido:2021idr`]; Espinosa-Portalés & García-Bellido, PDU 34 (2021) 100893, arXiv:2106.16012 [`Espinosa-Portales:2021cac`].

Quantum chaos, ETH, de Sitter holography — Maldacena, Shenker & Stanford, JHEP 08 (2016) 106, arXiv:1503.01409 [`Maldacena:2015waa`]; Cotler et al., JHEP 05 (2017) 118 [`Cotler:2016fpe`]; Page, PRL 71 (1993) 1291 [`Page:1993df`]; Deutsch, PRA 43 (1991) 2046 [`Deutsch:1991msp`]; Srednicki, PRE 50 (1994) 888, cond-mat/9403051 [`Srednicki:1994mfb`]; Rigol, Dunjko & Olshanii, Nature 452 (2008) 854 [`Rigol:2007juv`]; Banks, IJMPA 16 (2001) 910, hep-th/0007146 [`Banks:2000fe`]; Susskind, JHAP 1 (2021) 1, arXiv:2109.14104 [`Susskind:2021esx`]; Lin, JHEP 11 (2022) 060, arXiv:2208.07032 [`Lin:2022rbf`]; Narovlansky & Verlinde, JHEP 05 (2025) 032, arXiv:2310.16994 [`Narovlansky:2023lfz`].

Emergent gravity — Verlinde, JHEP 04 (2011) 029 [`Verlinde:2010hp`]; Verlinde, SciPost Phys. 2 (2017) 016 [`Verlinde:2016toy`].

Long-range systems, criticality, monitored dynamics — Campa, Dauxois & Ruffo, Phys. Rept. 480 (2009) 57, arXiv:0907.0323 [`Campa:2009jxa`]; Bak, Tang & Wiesenfeld, PRL 59 (1987) 381 [`Bak:1987xua`]; Hawking & Page, CMP 87 (1983) 577 [`Hawking:1982dh`]; Fisher et al., Ann. Rev. Cond. Mat. Phys. 14 (2023) 335, arXiv:2207.14280 [`Fisher:2022qey`].

Volume-law entanglement in cosmology — Khlebnikov & Sheoran, arXiv:1907.00487 [`Khlebnikov:2019wbk`].

Structure-sourced dark energy — Gough, *Information dark energy* (≈2008) [`Gough2008`]; Pandev, *Structural Entropy Dark Energy: a fixed-parameter, growth-gated holographic dark-energy model without a cosmological constant* (Paper I, companion to this work; in preparation, 2025) — the cosmology this paper's foundations underwrite.